



Improved humid air condensation heat transfer through promoting condensate drainage on vertically stripe patterned bi-philic surfaces

Junbeom Lee, Seunghwan Lee, Jaeseon Lee*

Innovative Thermal Engineering Laboratory, School of Mechanical Engineering, Ulsan National Institute of Science and Technology (UNIST), 50 UNIST Rd., Ulsan 44919, South Korea

ARTICLE INFO

Article history:

Received 6 April 2020

Revised 11 June 2020

Accepted 13 July 2020

Available online 24 July 2020

Keywords:

Condensation

Heat transfer

Heat exchanger

Bi-philic

Super-hydrophilic

Super-hydrophobic

Water harvesting

ABSTRACT

Even when dropwise condensation occurs on the super-hydrophobic surface, the condensed water droplets stick to the surface in the Wenzel state, resulting in lower drainage rate than the super-hydrophilic surface. Condensate that is not drained consequently degrades condensation heat transfer performance over time. In this study, it was considered that the striped patterned bi-philic surface could improve the condensation heat transfer in that droplet of the near super-hydrophobic area can be quickly removed and drained to a nearby super-hydrophilic region. Rapid drainage of condensate from the near super-hydrophobic surface can result in the renewal of new dropwise condensation, which can lead to increased heat transfer. Condensate discharges from bi-philic surfaces fabricated by material printing were measured and compared to condensate discharges from fully super-hydrophilic and near super-hydrophobic surfaces. As decreasing the stripe width of the bi-philic surface, the drainage rate is improved due to the rapid growth of the droplet nucleate on the near super-hydrophobic surface and the increment of the drainage path. The maximum condensate discharge from the bi-philic surface was measured to be 15% higher than that of the entirely super-hydrophilic surface. The amount of condensate recovery from the bi-philic surface was predicted through film and dropwise condensation combined modeling. Through this model, the amount of condensate recovered from the bi-philic surface was predicted within the 8% error range. Determination of the optimal stripe pattern to maximize the amount of condensate recovery can be made through developed modeling. The optimized stripe design of bi-philic surface resulted in a 25% increase in condensate discharge compared to the entirely super-hydrophilic surface.

© 2020 Elsevier Ltd. All rights reserved.

1. Introduction

Controlling the wettability of the surface can bring many advantages in terms of engineering applications. Depending on the surface wettability, the behavior of water droplets or condensate on the surface changes differently. Adjusting the surface wettability is inseparable from condensation heat transfer. When a solid surface is in contact with the surrounding moist air or vapor and the surface temperature is kept below the saturation temperature corresponding to the vapor pressure, condensation occurs where the liquid phase condenses on the surface.

Condensation can be classified into two types, film or dropwise condensation, depending on the form of condensate. In film condensation, the condensate on the surface spreads widely and forms a thin film through the coalescence process. In dropwise conden-

sation, the droplets remain partially spherical, and there is a relatively wide area where the heat transfer surface is not covered by condensate. Therefore, the heat transfer performance of film condensation, in which the liquid film acts as additional heat resistance, is inferior to that of dropwise condensation. There are many studies on the basic principles of the dropwise condensation because of the clear advantage in heat transfer [1–8]. Vemuri et al. [1] developed the theoretical modeling equation of dropwise steam condensation heat transfer, considering the droplet population density, and experimentally proved it. In this study, it was reported that the maximum droplet radius in the dropwise condensation greatly influences the heat transfer performance. Many studies have been published that evaluate the maximum condensate droplet radius at various surface wetting characteristics and investigate their relationship to heat transfer performance [9–14]. Recent developments in manufacturing technology have made it possible to quickly and efficiently fabricate near super-hydrophobic surfaces and create complex hierarchical structures using a variety of methods such as lasers [15–16]. Chemical etching is a sim-

* Corresponding author.

E-mail address: jaeseonlee@unist.ac.kr (J. Lee).

Nomenclature

A	Area [m^2]
b	Critical droplet size [m]
CA	Contact angle [deg]
C_p	Specific heat [$\text{J/kg}\cdot\text{K}$]
DWC	Dropwise condensation
FWC	Film condensation
g	Gravitational acceleration [m/s^2]
h_{fg}	Vaporization enthalpy [J/kg]
h_v	Convection heat transfer coefficient [$\text{W/m}^2\cdot\text{K}$]
k_l	Liquid thermal conductivity [$\text{W/m}\cdot\text{K}$]
$L_{interface}$	Total interfacial length on the bi-philic surface [m]
Pr	Prandtl number [-]
q	Heat flux [W/m^2]
Re	Reynolds number [-]
SAR	Super hydrophilic surface area ratio [-]
T_i	Gas-liquid interface temperature [K]
T_s	Vapor saturation temperature [K]
T_w	Surface temperature [K]
T_∞	Surrounding air temperature [K]
V	Water droplet volume [m^3]
W	Stripe pattern width [m]
Greek Symbols	
σ	Surface tension [N/m]
ρ_l	Liquid density [kg/m^3]
ρ_a	Air density [kg/m^3]
μ_l	Liquid dynamic viscosity [$\text{kg/m}\cdot\text{s}$]
θ_r	Receding angle [rad]
θ_a	Advancing angle [rad]
θ_s	Static contact angle [rad]
ω	Moisture fraction [-]

ple method to create hierarchical structures to adjust the wetting property [17]. Electrochemical jet processing (EJP) and high-current pulsed electron beam (HCPBE) provides an effective solution to control functional properties of nonwetting surfaces by creating hierarchical structures [18].

It is generally believed that hydrophobic surfaces with excellent water repellent properties will easily cause dropwise condensation and show better condensation performance. Much effort has been focused on the development of superhydrophobic surfaces. However, these hydrophobic surfaces may not always be optimal surface conditions for condensation in the humid air. When water droplets contact with hydrophobic surfaces from the outside, the water droplets and surfaces remain Cassie-Baxter state due to the low adhesion of the hydrophobic surfaces. However, another aspect is observed when condensation nuclei grow slowly on superhydrophobic surfaces through condensation. There is an observation that condensation nuclei, which initially adhere well to the surface, agglomerate with each other and move to Wenzel state rather than Cassie-Baxter state during the condensation process [19–25]. Since dropwise condensation is maintained, the heat transfer performance of the hydrophobic surface is still higher than that of the hydrophilic surface. However, the coalescence of condensate droplets eventually leads to the transition to the film condensation on the hydrophobic surfaces and degrades heat transfer performance. Methods have been proposed to prevent the transition to film condensation using a surface in the form of a combination of hydrophobic and hydrophilic surfaces. Some studies have made efforts to utilize these mixed bi-philic surfaces to increase the discharge of condensate and ultimately improve heat transfer [26–29]. Grooten et al. [29] studied the combined surface heat exchanger coated with Polyvinylidene Fluoride (PVDF) and titanium, which

show hydrophobicity and hydrophilicity, respectively. The droplet formed on the PVDF zone was quickly removed to the hydrophilic region due to the wet gradients, and it derived similar or higher heat transfer performance compared with the entire PVDF surface.

Studies have also been conducted to analyze the condensation characteristics of the bi-philic patterned surfaces according to flow conditions. Yang et al. [30] compared the heat transfer coefficient of super-hydrophilic, hydrophilic, hydrophobic, and bi-philic surfaces with changes in air velocity and relative humidity. A bi-philic surface of V-shaped channel design combined with vertical hydrophilic channel served as a drainage trough led to quick droplet removal. By using the bi-philic surface, the heat transfer coefficient is enhanced by 3 - 9% compared to the hydrophobic surface and 6 - 16% compared to the hydrophilic surface. In addition to experimental studies, numerical analyses on bi-philic surfaces have also been performed. Peng et al. [31] studied the heat transfer mechanism of stripe patterned bi-philic surface under the steam condensation conditions. The heat flux of the hydrophobic surface was derived by considering the maximum droplet radius, and the hydrophilic surface heat flux could be obtained by inducing liquid film thickness using the mass conservation of condensate droplets grown on the entire surface. Then, through the sum of the surface heat fluxes, the optimum heat flux was obtained by calculating the total heat flux and changing parameters such as subcooling and maximum droplet radius. Further studies have validated their model experimentally [32].

Most researches on the near super-hydrophobic surface and bi-philic surface up to now focused only on how to maintain dropwise condensation, which has the advantage of heat transfer. However, as long as the removal of the condensate droplets from the dropwise condensation surface is not performed efficiently, the improvement of the heat transfer performance is limited. In this respect, the present study focused on an approach to improve the overall condensation heat transfer by using partial hydrophilic surface areas as condensate drains on the vertical surface. It is meaningful that the condensed water condensed in the hydrophobic zone is quickly discharged through the hydrophilic region so that the dry state of the hydrophobic region can be maintained. In this approach, it is crucial to improve condensate discharges from the condensation surface. Mahapatra et al. [33] used a similar approach to enhance condensate drainage through bi-philic surfaces, but no quantitative analysis to characterize the moisture-collection was taken.

In this study, we attempted to find a way to maximize the drainage of condensate, taking into account the simple repetitive bi-philic stripe pattern to rule out the effects of complex geometry parameters. With changing the near super-hydrophobic and super-hydrophilic stripes spacing and super-hydrophilic area ratio deliberately, variations in collected droplets mass and condensate behavior are discussed in detail. A method of improving condensation heat transfer was sought through a comparative analysis of the condensate recovery from the bi-philic surface with entirely super-hydrophilic and near super-hydrophobic surfaces. This study proposes a method for determining the optimal stripe pattern for condensate drainage and maximizing condensation heat transfer by considering the fabrication method of bi-philic surfaces on a large metal surface.

2. Test sample preparation

2.1. Fabrication of bi-philic surfaces

Aluminum alloys (Al6061) are commonly used on heat transfer surfaces such as heat exchangers and their fins. Al6061 was used to fabricate three sample surfaces having different wettability, including super-hydrophilic, near super-hydrophobic, and bi-philic

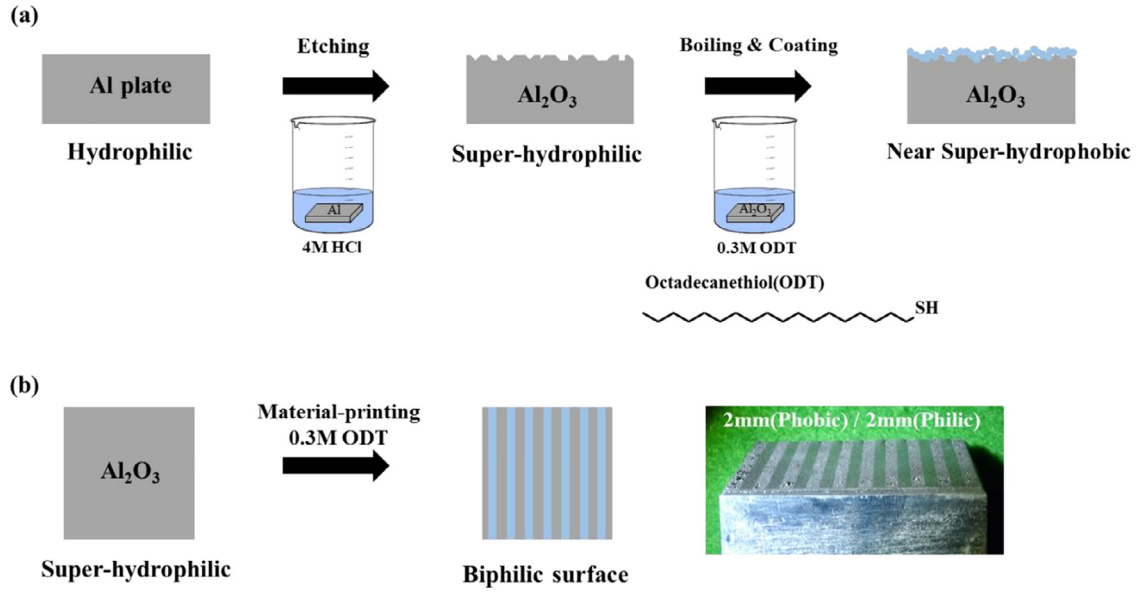


Fig. 1. The fabrication process of (a) Super-hydrophilic, near super-hydrophobic, and (b) bi-phobic surfaces.

surfaces in this study. The prepared $40\text{ mm} \times 40\text{ mm} \times 15\text{ mm}$ sized Al6061 blocks were polished using P320 and P1200 sandpaper before further process applications to maintain a constant initial surface roughness. Samples were cleaned in ultra-sonicator for 5 min in ethanol and 10 min in DI-water after surface polishing to remove the oxide layer and impurities on aluminum surfaces. Fig. 1(a) shows the fabrication process of super-hydrophilic and near super-hydrophobic surfaces. The rinsed Al block was etched in a 4 M HCl solution for 15 min to make a super-hydrophilic surface. The etched sample can quickly react with gasses in the air and lose their super-hydrophilic properties over time. Therefore, it was immersed in boiling DI water for 8 min to maintain the super hydrophilic surface properties [34]. To make the entire surface near super-hydrophobic, the prepared super-hydrophilic surface was immersed in 0.3 M Octadecanethiol (ODT) solution of the ethanol solvent at 50 °C for 1 hour. Through dip coating, the self-assembled monolayer of ODT which facilitates near super-hydrophobicity with low surface energy is formed. Fig. 1(b) represents a bi-phobic surface in which super-hydrophilic and near super-hydrophobic surfaces are repeatedly patterned in a stripe form. The near super-hydrophobic areas were printed on a super-hydrophilic base using a material printer (Dimatix Materials Printer DMP-2850). 0.3 M ODT solution is ejected with 10 pL cartridge from four nozzles of printhead. The inkjet printing process is repeated three times to form a stable and durable near super-hydrophobic pattern. The time taken to make a bi-phobic surface sample was about 1 hour.

The measured DI-water droplet contact angles on super-hydrophilic and near super-hydrophobic surfaces were shown in Fig. 2. Contact angles were measured using SEO Phoenix 300. The static angle of the super-hydrophilic surface is 4°, and that of the near super-hydrophobic surface is 142°. Static angle, advancing angle and receding angle for each surface are also presented in Fig. 2.

The SEM images in Fig. 3 show the difference in details of the prepared surfaces in this study. The number at the top of each SEM images indicates the magnification. As shown in the high magnification images of Fig. 3(a) and 3(b), it is observed that the needle-like microstructure of the ODT polymer is coated on the entire surface. Such coated microstructures result in the transition of super-hydrophilic surfaces to near super-hydrophobic surfaces. Fig. 3(c) represents the SEM image of the stripe patterned bi-phobic surface.

There is a difference in brightness between the super-hydrophilic and near super-hydrophobic zones, and it is possible to confirm that the stripe pattern is repeated. It can be seen that a smaller amount of ODT was applied to the near super-hydrophobic region by printing than the immersed coating, but needle-like microstructures are still observed in the near super-hydrophobic area.

2.2. Determination of bi-phobic surface stripe width

The bi-phobic surface has a longitudinal stripe pattern, and it was composed of super-hydrophilic and near super-hydrophobic surfaces repeatedly. The intended role of the super-hydrophilic surface is to drain the condensate, and it expects the drainage improvement by absorbing nearby droplets grown on the near super-hydrophobic surface. When the bi-phobic surface was prepared, the minimum stripe width was determined by considering the critical droplet size of the near super-hydrophobic surface. The critical droplet size means the minimum departure droplet radius when the surface tension force and gravity force of condensate droplet become balanced at the tilted surface. If the stripe width is smaller than the critical droplet size, condensate droplets on the near super-hydrophobic surface are not drained through the super-hydrophilic region, and interference may occur between the stripes. The critical droplet size on the vertical plate can be calculated through the following equations [12–13],

$$\frac{24}{\pi^3} \sigma b (\cos\theta_r - \cos\theta_a) = \rho_l g V \sin\alpha \quad (1)$$

$$V = \frac{\pi b^3}{24} \left(\frac{2 - 3\cos\theta_{avg} + \cos^3\theta_{avg}}{\sin^3\theta_{avg}} \right) \quad (2)$$

In the Eq. (1) and Eq.2, b means the critical droplet diameter, and θ_s , θ_r and θ_a indicate the static contact angle, receding angle, and advancing angle, respectively. The contact angle measured in Fig. 2 was applied to calculate a critical droplet diameter of about 0.5 mm at the near super-hydrophobic surface. Based on this value, the near super-hydrophobic width (W_{dwc}) was determined, and the stripe width of the sample was determined by fixing the area ratio of the entire super-hydrophilic surface area ratio (SAR). A total of 25 bi-phobic surface samples were made based on 50%, 60%, 75%, and 85% SAR. The SAR means the ratio of the super-hydrophilic

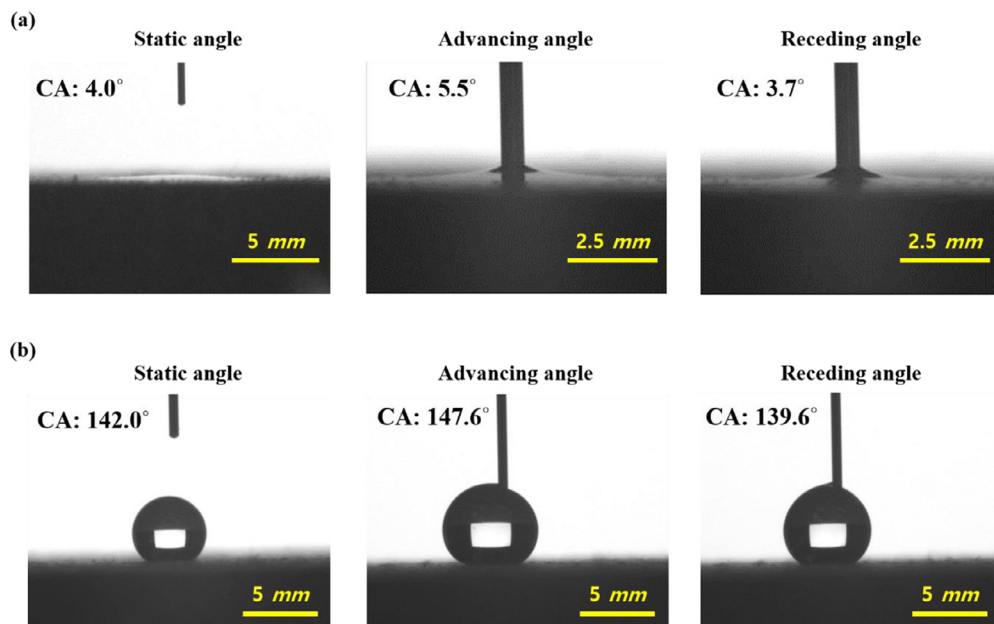


Fig. 2. Contact angles of water droplet on (a) super-hydrophilic, and (b) near super-hydrophobic surfaces.

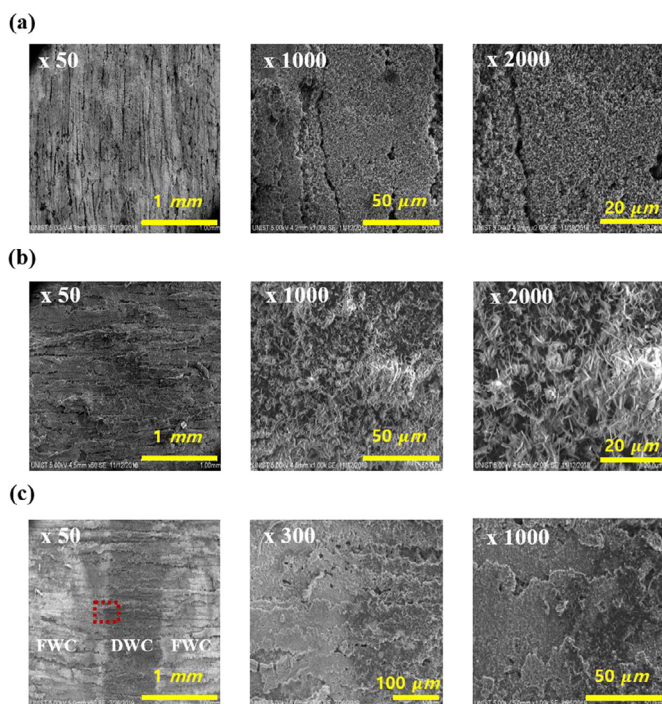


Fig. 3. SEM images on (a) super-hydrophilic, (b) near super-hydrophobic, and (c) bi-philic surfaces.

area to the total area of the bi-philic surface. The information of each sample is summarized in Table 1.

3. Experimental methods

Fig. 4 shows the experimental setup and test section for measuring the amount of condensate drainage on the test surface with humid air. The entire experimental setup consists of two loops. The first loop is a wind-tunnel section for moist air supply and condensate drainage measurement. The other is water-loop for controlling the temperature of the test surface. The dimension of the wind-

Table 1

Width of both near super-hydrophobic and super-hydrophilic stripe in bi-philic surface samples (W_{dwc} is the width of the near-superhydrophobic region and W_{fwc} is the width of the superhydrophilic region, SAR is super hydrophilic surface area ratio).

Sample No.	W_{dwc} [mm]	W_{fwc} [mm]	SAR [%]
1 - 5	1.50 - 2.60	1.50 - 2.60	50
6 - 13	0.64 - 2.70	0.96 - 3.90	60
14 - 21	0.55 - 2.50	1.65 - 7.50	75
22 - 26	0.60 - 1.50	3.40 - 8.50	85

tunnel is 14.5 cm x 14.5 cm x 110 cm and is made of acrylic. An axial fan and humidifier were installed at the entrance to induce the airflow and to supply moisture. The temperature and relative humidity of the inlet air were measured by a type-T thermocouple and hygrometer placed in the mid of wind-tunnel. A nozzle cone and airflow meter were installed at the end of the wind tunnel to increase the mixing degree of the outlet air and measure the average flow velocity of the air. The fully saturated 100% relative humidity was maintained by the additional humidifier, and the inlet air temperature was maintained at 14 °C to 15 °C. The flow velocity of the air was maintained at a fixed 0.6 m/s. Measurement errors of the airflow meter and thermocouples are 1% of full scale and ± 0.5 °C. The temperature of the entire water-loop was controlled by a water bath. The water inside the loop was circulated by using a micro-gear-pump. The inlet temperature of the cooling water was maintained at 5 °C.

Although not clearly shown in the test section structure of Fig. 4(b), the test surface and cooling plate were attached with carbon tape to reduce contact resistance. The side of the test surface and cooling plate were insulated to reduce the heat loss. The condensate water from the test surface was collected by a reservoir bucket at the bottom of the test section and weighed by a scale. Fig. 4(c) represents a picture of the experimental apparatus in this study. The total time of the experiment was 4 h for each sample. Condensate measurements were taken every 30 min after reaching steady-state about 1 hour. The condensate droplet behavior on the test surface was recorded using a digital macro camera. The signals from the thermocouple, airflow meter, and hygrometer are

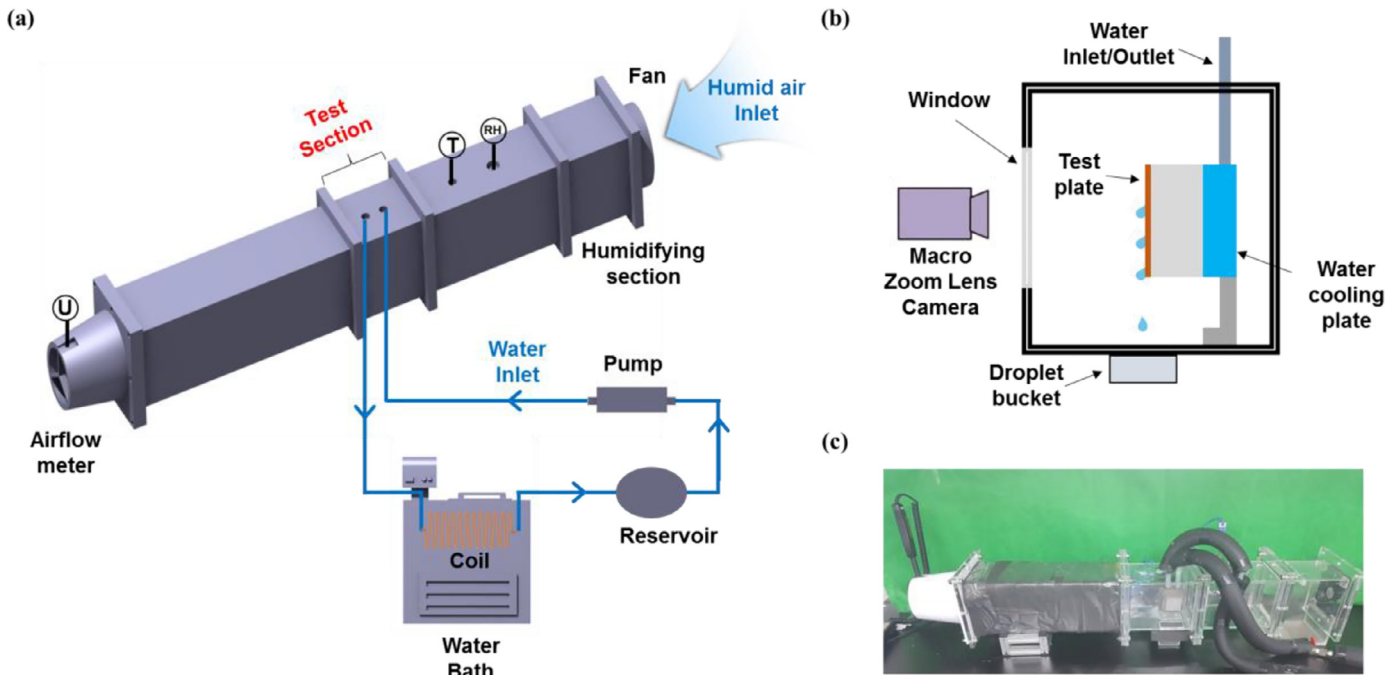


Fig. 4. (a) Schematic of an experimental setup for humid air condensation, (b) cross-sectional view of the test section, (c) photograph of experimental apparatus.

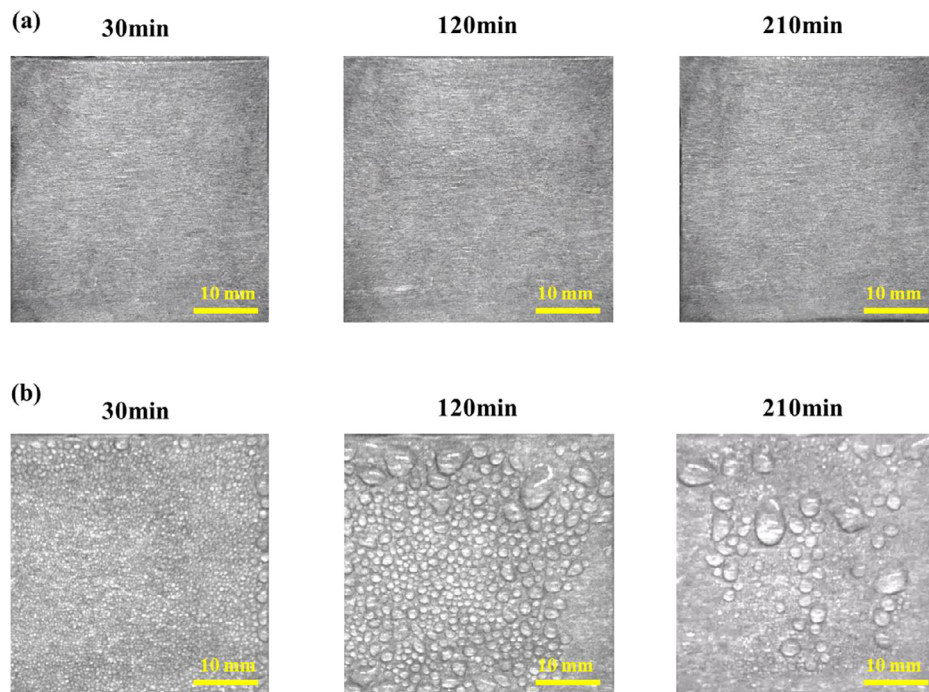


Fig. 5. Comparison of moisture condensation in humid air on (a) super-hydrophilic and (b) near super-hydrophobic surfaces over time (the inlet air temperature maintained at 14 °C, the relative humidity is 100%, and the inlet temperature of the cooling water is 5 °C).

transferred to the NI-DAQ module and processed by the data acquisition system.

4. Results and discussions

Fig. 5 shows the condensation processes on the surfaces at 30, 120, and 210 min after the start of the test on the super-hydrophilic and near super-hydrophobic surfaces. As shown in Fig. 5(a), the initially formed condensation nuclei are easily combined to form a thin water film over the entire surface in

the super-hydrophilic surface. Over time, the condensate film, which flows down, builds up thickly on the lower part of surfaces by gravity. It is difficult to find a difference in condensation behavior over time because droplets do not appear on the super-hydrophilic surface where the film condensation proceeds. As shown in Fig. 5(b), relatively small condensate droplets were formed evenly over the entire surface in the near super-hydrophobic surface of the initial stage of condensation. However, the projected cross-sectional shape of the initial small droplets or of the larger droplets merged over time (120 min) is not an ex-

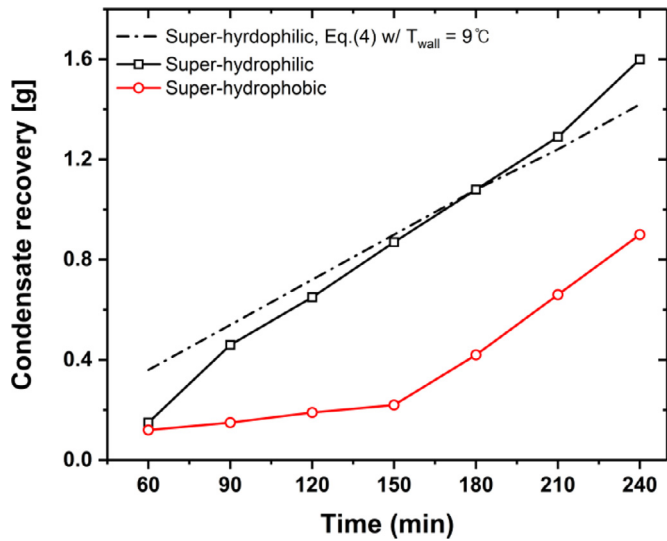


Fig. 6. Condensate water recovery amount of the homogeneous surfaces over time (the inlet air temperature maintained at 14°C , the relative humidity is 100%, and the inlet temperature of the cooling water is 5°C).

act circle. The non-circular base shape means that condensation droplets grown from near super-hydrophobic surfaces have strong adhesion by the surface due to the contact area increase of the droplet base. Therefore, it can be judged that these droplets remain in the Wenzel state, and the separation of the droplets from the surface is not easy. Over time these droplets grow larger and then overcome the adhesion of the surface and fall off by gravity. Large falling drops swept through the small droplets along the path, leaving the remaining moisture on the surface. These residual waters lead to further condensation progression to film condensation, which eventually causes the droplet condensation to fade over time, even on near super-hydrophobic surfaces. The result shows that the dropwise condensation is not well maintained, and there is no enhancement in condensation heat transfer on near super-hydrophobic surfaces where dropwise condensation is expected.

Fig. 6 shows the amount of condensate recovered from the homogeneous super-hydrophilic and near super-hydrophobic surfaces over time. Measurements were recorded one hour after reaching the steady-state of the condensation process. The rapid coalescence between condensate droplets on the super-hydrophilic surface leads to the formation of a water film over the entire area, leading to faster drainage of condensate. The amount of condensate on the super-hydrophilic surface increases linearly over time, which is consistent with the previous report [35]. As observed in Fig. 5(b), dropwise condensation occurs in the initial stage on the near super-hydrophobic surface, but condensate drainage does not quickly occur due to the presence of droplets in the Wenzel state. About 150 min after the start of the test, the discharge of condensate from the near super-hydrophobic surface begins in earnest (Fig. 6). The condensation rate rises as the droplets grown faster on the upper edge start sweeping after 150 min of the start. This is the time estimated after the initial droplets combined with an increase in size and the transition to film condensation occurred in significant portions. Consequently, the discharge of condensate from the near super-hydrophobic surface is much less than the amount from the super-hydrophilic surface. Condensate drainage from the film condensation surface can be predicted by a theoretical model (Lee et al. [12]). The calculation by this theoretical model and the current experimental results are compared in Fig. 6.

The dotted black line represents the value calculated by the theoretical model.

Similar to Nusselt's film condensation analysis, this model consists of a steady-state energy conservation equation consisting of humid air convection, latent heat when moisture condenses, and conduction through a liquid film. Each heat flux term was described from Eq. (3.1) to Eq. (3.4) [12].

$$q_v + q_L = q_d \quad (3.1)$$

$$q_v = h_v(T_\infty - T_i) \quad (3.2a)$$

$$Nu_x = \frac{0.3387 Re_x^{1/2} Pr^{1/3}}{\left[1 + \left(\frac{0.0468}{Pr}\right)^{2/3}\right]^{1/4}} \quad (3.2b)$$

$$\overline{Nu}_x = 2Nu_x \quad (3.2c)$$

$$q_L \approx \frac{h_v h_{fg}}{c_p} (\omega_\infty - \omega_i) \quad (3.3)$$

$$\omega_j = 3.7444 \times 10^{-3} + 0.3078 \times 10^{-3} T_j + 0.0046 \times 10^{-3} T_j^2 + 0.0004 \times 10^{-3} T_j^3 \quad (3.3a)$$

$$q_d = \frac{k_l}{\delta} (T_i - T_w) \quad (3.4)$$

$$\delta \approx \left[\frac{4k_l \mu_l (T_s - T_w) L}{g \rho_l (\rho_l - \rho_a) h_{fg}} \right]^{1/4} \quad (3.4a)$$

The term q_v , q_L and q_d represents the heat flux related to air convection, phase change during the condensation, and conduction in the liquid film, respectively. For a term of air convection, h_v , the forced convection equation developed by Churchill (1976) and Rose (1979) [36,37] was used (Eq. (3.2(b)) – Eq. (3.2(c))). Finally, the theoretical amount of condensate drainage from the super-hydrophilic surface can be calculated as follows,

$$\dot{m} = \frac{q_L A}{h_{fg}} \quad (4)$$

Term A indicates the entire area of the test surface. The slopes of the condensate quantity over time in the experimental and theoretical results are 0.400 g/h and 0.355 g/h , respectively. The reason for the difference between the two values is presumed to be because the effect of forced convection is not reflected in the original model and geometric factors, such as the relatively long edge effect [38] to the total condensation surface area.

Fig. 7 shows the progress of condensation on the bi-philic surfaces at super-hydrophilic area ratios of 50%, 60%, 75%, and 85%. It can be continuously observed that dropwise condensation occurs in the near super-hydrophobic area, and the film condensation occurs in the super-hydrophilic area. It has been observed that droplets grown in the near super-hydrophobic region are absorbed and discharged into the super-hydrophilic region before reaching the critical droplet size. Also, even if the droplets of the near super-hydrophobic region grow to critical droplet size, they are absorbed into the super hydrophilic region due to the strong wettability gradient between the super hydrophilic region and the near super-hydrophobic region rather than falling by gravity. When the super-hydrophilic stripe width is much smaller than the critical droplet size (Fig. 7(c)), a phenomenon is observed in which condensate droplets merge onto the superhydrophobic region. It is believed that these droplets are maintained with a surface adhesion greater than the gravitational force acting on the combined droplets at a given width. As a result, condensate droplets

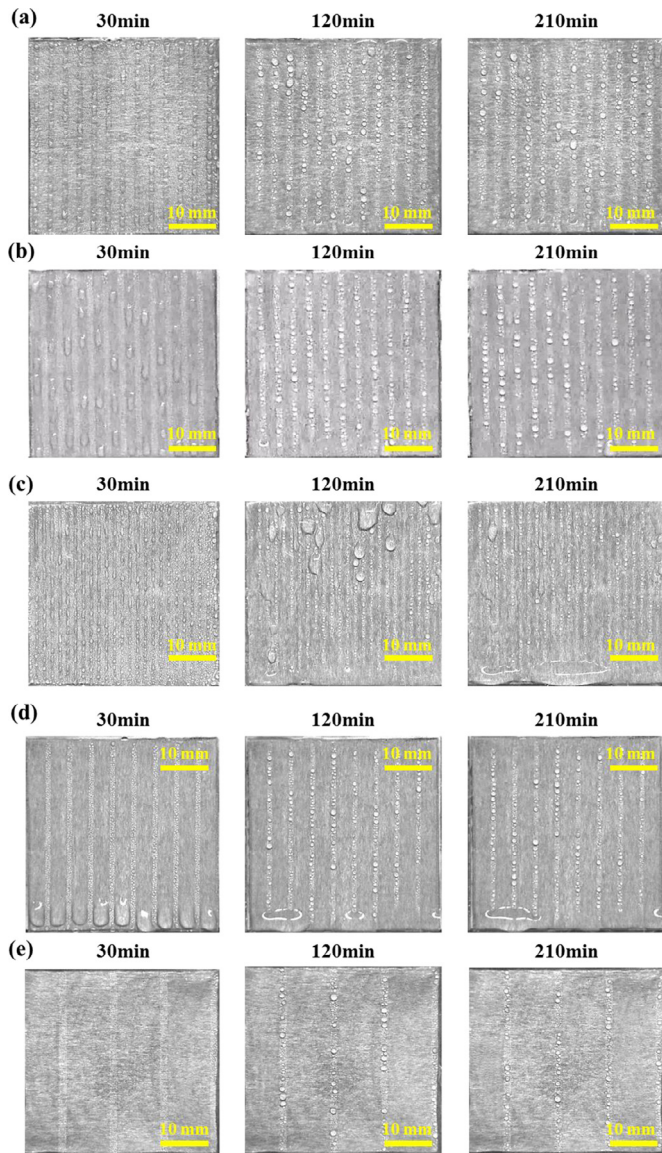


Fig. 7. Comparison of condensation characteristics on stripe pattern bi-philic surface over time; (a) $W_{dwc} = 1.5$ mm, $W_{fw} = 1.5$ mm, SAR = 50%, (b) $W_{dwc} = 1.3$ mm, $W_{fw} = 2$ mm, SAR = 60%, (c) $W_{dwc} = 0.64$ mm, $W_{fw} = 0.96$ mm, SAR = 60% (d) $W_{dwc} = 1.1$ mm, $W_{fw} = 3.4$ mm, SAR = 75%, (e) $W_{dwc} = 1.5$ mm, $W_{fw} = 8.5$ mm, SAR = 85%. (the inlet air temperature maintained at 14 °C, the relative humidity is 100%, and the inlet temperature of the cooling water is 5 °C).

on the super-hydrophilic region cover the adjacent near super-hydrophobic regions. This phenomenon is called 'bridge droplet'. The bridge droplets were drained as the droplet grows further, but it adversely affects drainage and heat transfer performance because the droplets cover a near super-hydrophobic region. The condensate collected in the super-hydrophilic region flows downward due to gravity, which produces an edge effect that agglomerates before being completely drained at the end of the surface. Some of the liquid films at these ends form bridges again and merge, hindering condensation heat transfer in the near super-hydrophobic region (Fig. 7(c), (d)). In the case of Fig. 7(e), the bridging phenomenon of such a film is reduced because the width of the super-hydrophilic region is wide.

Fig. 8 shows the time-dependent change of the amount of condensate recovery over the entire test surfaces, according to SAR. Condensate recovery at each SAR surface was compared to near super-hydrophobic and super-hydrophilic surfaces. After a certain

time, the amount of condensate recovery on the bi-philic surface increases linearly as a whole. It also shows that the condensate amount from the bi-philic surface is more than the near super-hydrophobic surface due to the improved drainage.

As the near super-hydrophobic stripe width increased on all bi-philic surfaces, the amount of condensate decreased. Increasing the near super-hydrophobic width causes the number of near super-hydrophobic and super-hydrophilic stripe pairs to decrease. As a result, the total length of the interface boundary of the two regions, which is a path for droplets condensed in the near super-hydrophobic area to move to the super-hydrophilic area, is shortened. The positive thing about the test results is that, as shown in Fig. 8, some test surface cases were observed where more of the mass of condensate is recovered than super-hydrophilic surfaces. The maximum enhancement case of the drainage is about 15% more compared with the super-hydrophilic surface. Also, the total condensate mass increases with the increase in the super-hydrophilic area, i.e., as the SAR increases. However, when comparing the Fig. 8(c) and (d), it can be confirmed that the amount of condensate recovery does not continuously increase with increasing SAR, and there is an optimized SAR value that can maximize it. Further explanations will be provided regarding this trend in the next paragraph.

Fig. 9 shows the amount of condensate recovery on the repeated stripe pattern bi-philic surfaces according to the change in super-hydrophobic width for each SAR. For comparison, the recovery amount of the entirely super-hydrophilic surface is shown in solid lines, and the recovery amount per individual area according to the arithmetic ratio of each SAR is indicated by a dotted line. The bar in the hatch pattern indicates the whole recovered condensate mass from the bi-philic surface. The value indicated by the blue bar means a value obtained by subtracting the amount of the super-hydrophilic recovery multiplied by the ratio of SAR from the total amount of condensate recovery on the bi-philic surface. The lower dashed line calculates the recovery from a single superhydrophobic area according to the SAR ratio. The results are displayed in this way because it is thought that the comparison of the near super-hydrophobic area recovery (lower dashed line) with the blue bar value may indicate the degree of improvement in dropwise condensation.

The increase in the near super-hydrophobic area in the stripe pattern, i.e., the reduced SAR and the wide near super-hydrophobic stripe width, results in a reduction in condensate recovery. For relatively large SAR and narrow near super-hydrophobic stripe widths, the condensate recovery performance is much improved. In these test cases, the blue bar data showed a value far exceeding the lower dashed line. These data also imply that improvements in condensate recovery over the entire bi-philic surface have been achieved, particularly with the increased recovery of condensate over the near super-hydrophobic area. As described above, when the near super-hydrophobic stripe width is reduced, the length of the drainage path leading the droplets to the super-hydrophilic zone increases. It can be concluded that the dropwise condensation proceeds rapidly on the surface again, and the overall heat transfer of condensation improves as the droplets are quickly discharged in the near super-hydrophobic region.

Fig. 10 compares the recovery of condensate collected over 4 h from all test surface samples. The measurement of the condensate recovery was started an hour after condensation being initiated. The exclusion of the initial period experiment was intended to minimize the effects of transient conditions [39]. Both ends of the x-axis show the data on the super-hydrophilic and near super-hydrophobic surfaces, respectively, and the bi-philic stripe pattern surfaces are indicated by the near super-hydrophobic and super-hydrophilic widths for each SAR. As shown in the previous test results, it can be confirmed that the overall condensate recovery

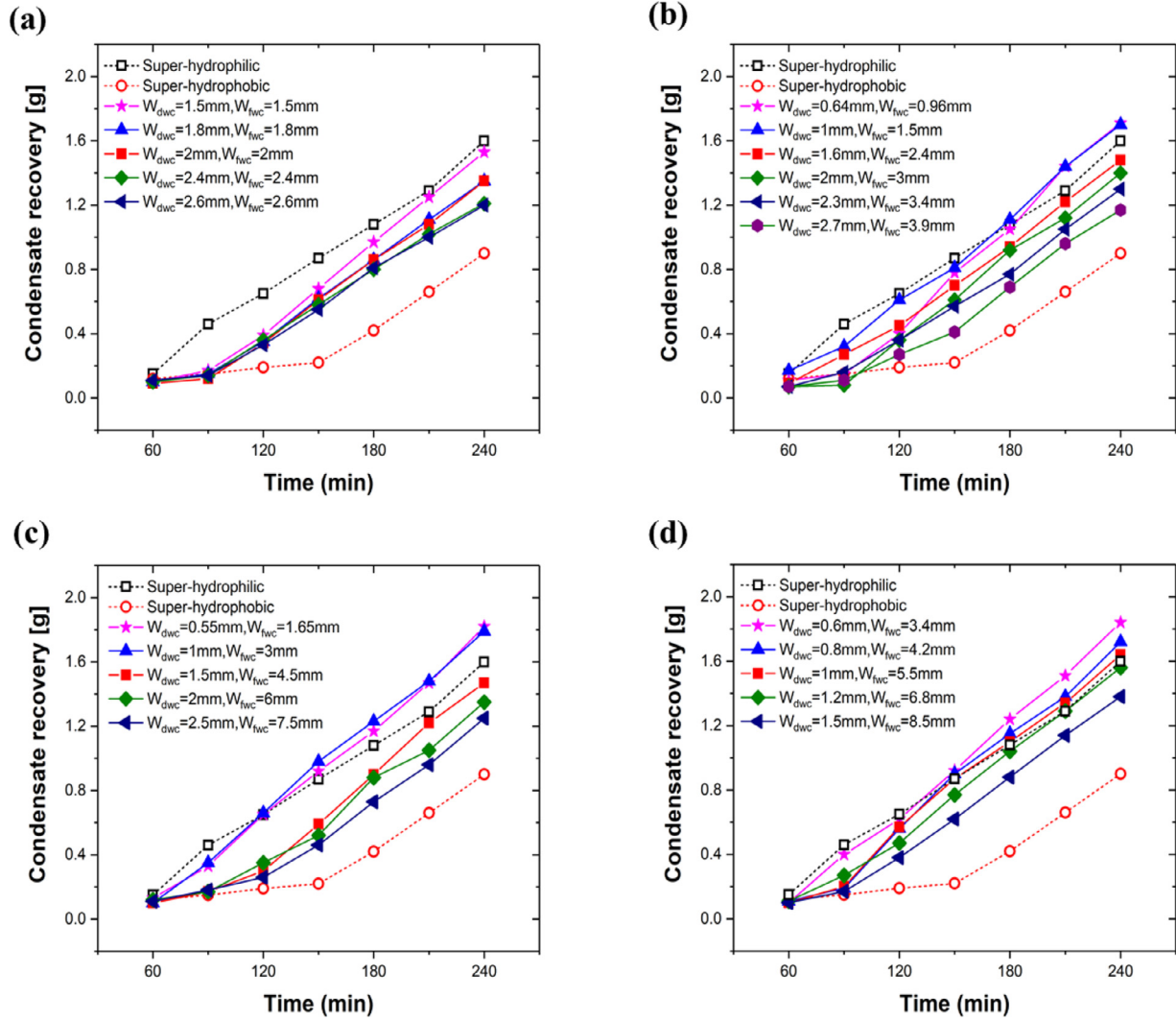


Fig. 8. Condensate mass change over time according to the area ratio of super-hydrophilic region, (a) SAR=50%, (b) SAR=60%, (c) SAR=75%, (d) SAR=85% (the measurement error of recovery weight is ± 0.1 g).

tends to increase with increasing SAR. Also, as the width of the near super-hydrophobic pattern in each SAR data becomes narrower, the amount of condensate recovery increases. However, it can be confirmed that there is a threshold for this increase. The stagnant increase in the amount of condensate recovery after the threshold point is related to the merging of the 'bridge droplet' mentioned in Fig. 7(c). From the overall trend of the results of these experiments, it is possible to predict the condensate recovery amount according to changes in stripe pattern geometric parameters.

The condensate recovery of the bi-philic surface is ultimately expressed as the sum of film condensation from the super-hydrophilic surface and improved dropwise condensation from the near super-hydrophobic surface. For the contribution of the film condensation, the area ratio of the super-hydrophilic surface (SAR) is multiplied by Eq. (4) and expressed by the following equation,

$$\dot{m}_{phil} = \frac{q_L \times A_{total}}{h_{fg}} \times \frac{A_{hydrophilic}}{A_{total}} (SAR). \quad (5)$$

The improved amount of dropwise condensation on the near super-hydrophobic surface needs to be expressed as a single geometric variable representing a complex change in the stripe pattern. The length of the interface between the hydrophilic and hydrophobic regions can be considered as a variable representing

such bi-philic stripe characteristics. Fig. 11(a) shows the calculated condensate amount from the near super-hydrophobic area only according to the change in the interfacial length. The amount of recovered condensate on the graph y-axis is equal to the value indicated by the blue bar in Fig. 9. It can be confirmed that the amount of condensate recovery from the near super-hydrophobic surface shows a particular trend depending on the interface length. A single curve can fit this trend and Eq. (6) represents the fitting. The regression coefficient R^2 value is 0.991. From the fact that the regression coefficient R^2 value is sufficiently high, it can be seen that this fitting curve accurately fits the experiment data by Eq. (6).

$$\dot{m}_{phob} = 0.695 * (1 - e^{-4.2488 * L_{interface}})^{6.744} \quad (6)$$

Then, by combining Eq. (5) and Eq. (6), the recovery of condensate from the entire bi-philic surface can be estimated as follows,

$$\dot{m} = \dot{m}_{phil} + \dot{m}_{phob}. \quad (7)$$

The comparison of the measured value of the amount of condensate recovery from the bi-philic surface with the predicted value by Eq. (7) is presented in Fig. 11(b). It can be confirmed that the tendency and amount of condensate from the bi-philic surface is predicted accurately according to the interfacial length and SAR change. As the interfacial length decreases, the absorption

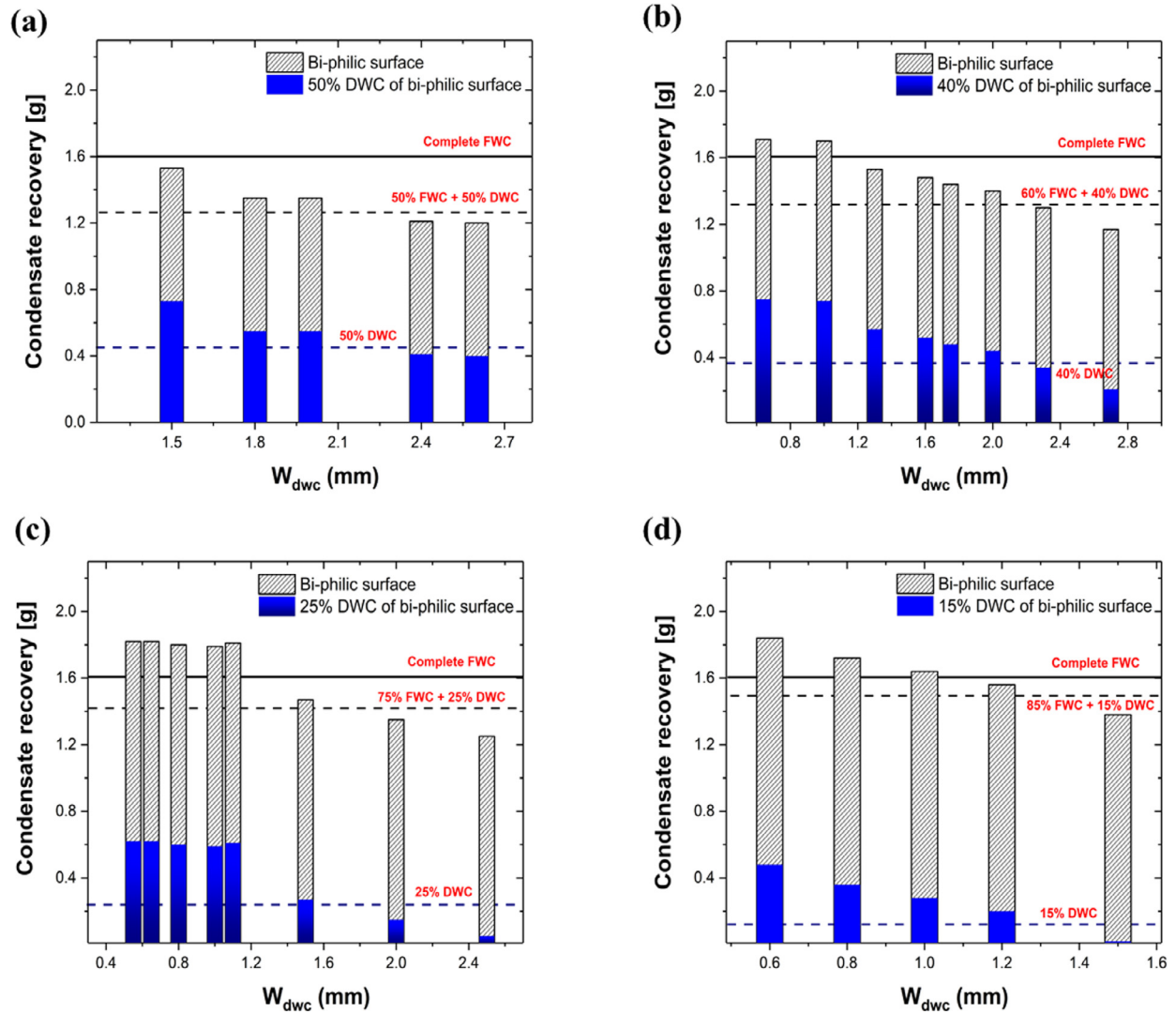


Fig. 9. Comparison of 4 hour condensate recovery for each surface according to changes in DWC width, (a) SAR = 50%, (b) SAR = 60%, (c) SAR = 75%, (d) SAR = 85%.

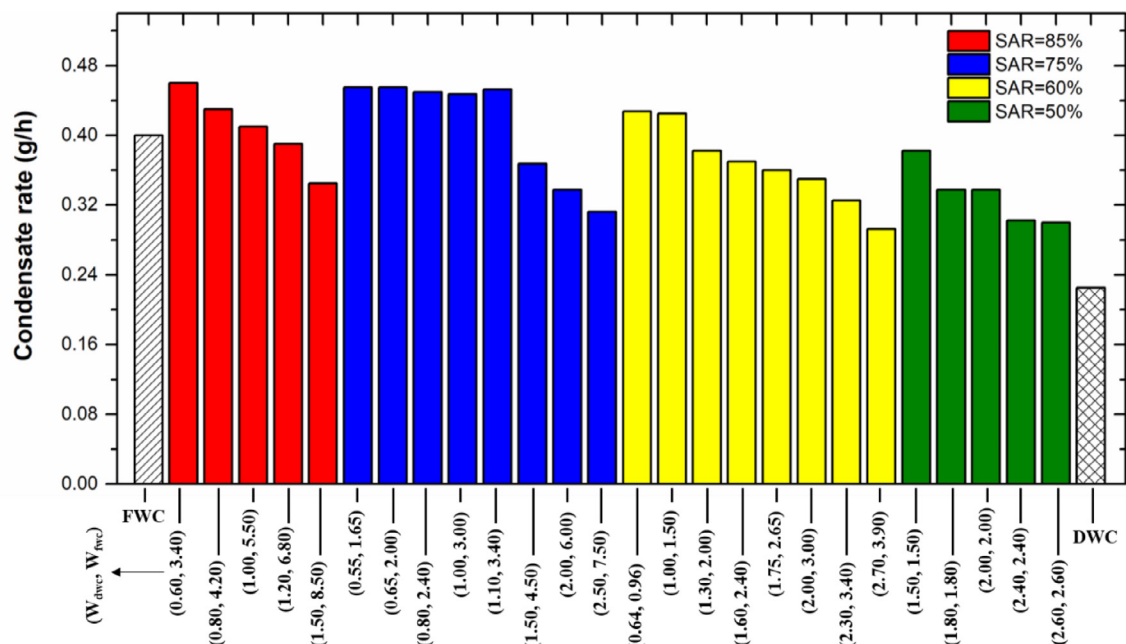


Fig. 10. Comparisons the condensate water recovery amount from the test surfaces.

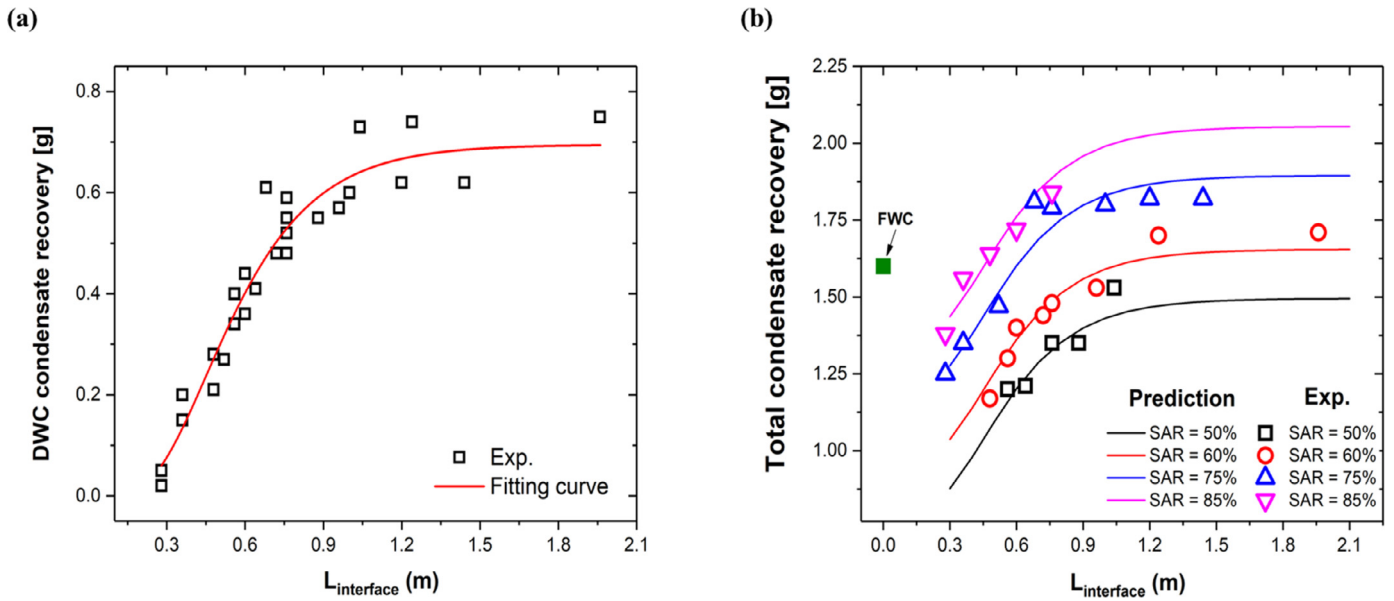


Fig. 11. (a) The amount of condensate recovered only from the near super-hydrophobic area within the bi-philic surfaces, (b) comparison of the amount of condensate recovered from the entire bi-philic surfaces and the predicted value by Eqn. (7). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

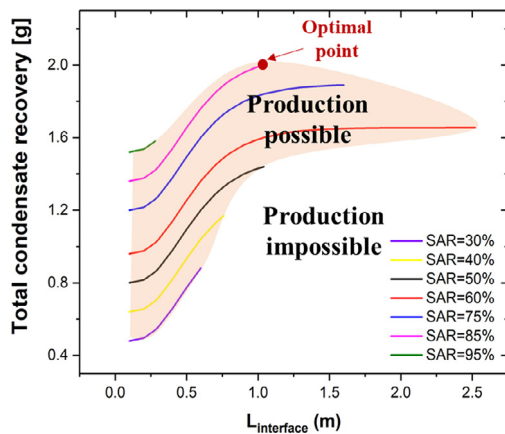


Fig. 12. Prediction of bi-philic surface condensate mass according to changes in hydrophilic area ratio (the contact angle of super-hydrophilic surface is about 4° and near super-hydrophobic surface is about 142°). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

of droplets decreases due to a decrease in the path from the near super-hydrophobic surface to the super-hydrophilic surface, which results in a reduction of the total amount of bi-philic surface condensate recovery. The fabrication of the stripe pattern of the printing method used in this study is not possible in all SAR ranges due to limitations in geometric characteristics and resolution. By simultaneously expressing the range of geometric parameters that can be fabricated and Eq. (7), the optimal pattern for maximum condensate recovery can be obtained.

Fig. 12 shows the range that can be fabricated for each SAR, and the predicted value of the amount of condensate recovered within the range. There are areas where all $L_{interface}$ ranges cannot be manufactured due to limitations in geometric characteristics and print resolution. Overall, when $L_{interface}$ increases, the amount of the total condensation recovery increases due to the interaction of the super-hydrophilic and near super-hydrophobic regions. When SAR

is 50 to 85%, it is observed that the total condensation rate tends to stop increasing as $L_{interface}$ increases.

The conditions for obtaining the maximum amount of condensate recovery in the shaded area are 85% SAR when the width of the super-hydrophilic region (W_{FWC}) is 2.8 mm, and the width of the near super-hydrophobic region (W_{DWC}) is 0.5 mm. Under the optimal pattern conditions, the predicted amount of condensate is about 2 g, and this value is improved by 25% compared to an entirely super-hydrophilic surface. Among the measured bi-philic samples, the closest to the stated optimum was $W_{FWC} = 3.4$ mm, and the $W_{DWC} = 0.6$ mm. The amount of condensate recovery is 1.84 g. The amount of condensate recovery was slightly smaller than the predicted maximum. It can be expected that the improved condensate recovery was all due to the rapid removal of droplets from the superhydrophobic surface and that the overall condensation heat transfer was improved. Experimental measurement studies on the improvement of condensation heat transfer have difficulty in measuring a small amount of temperature gradient due to low heat flux in a small test surface area. For this reason, it is planned as the next step of research.

5. Conclusions

The feasibility of improving condensation heat transfer in humid air using a bi-philic surface was examined. Bi-philic surface with repetitive stripe patterning was fabricated by material printing to improve the drainage on the near super-hydrophobic surface. It was experimentally confirmed that the droplets condensed in the near super-hydrophobic area moved to the super-hydrophilic region along with the interface, and the drainage of the film condensation increased. Based on the data of test surfaces, it was possible to predict the condensate amount from the bi-philic surface considering the increase in dropwise condensation due to the increase in the interfacial length, and this prediction can determine the stripe pattern that maximized the discharge of condensate. **The maximum amount of condensate on the bi-philic surface occurred at a super-hydrophilic area ratio of 75% or more.** The maximum amount of condensate measured in the experiment is about 1.84g, which is 15% greater than that of a fully super-hydrophilic surface. The maximum condensate discharge obtained through the

prediction could be expected to increase more than this. As a further study, more experiments will be needed to derive the optimal stripe patterning structure, considering not only drainage enhancement but also heat transfer performance.

Declaration of Competing Interest

None.

Acknowledgement

This material is based upon work supported by the research funding from the Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (Grant No. NRF-2017R1D1A1B03029858).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijheatmasstransfer.2020.120206](https://doi.org/10.1016/j.ijheatmasstransfer.2020.120206).

References

- [1] S. Vemuri, K.J. Kim, An experimental and theoretical study on the concept of dropwise condensation, *Int. J. Heat Mass Transf.* 49 (3) (2006) 649–657.
- [2] M. AbuOrabi, Modeling of heat transfer in dropwise condensation, *Int. J. Heat Mass Transf.* 41 (1) (1998) 81–87.
- [3] J.L. McCormic, E. Baer, On mechanism of heat transfer in dropwise condensation, *J. Colloid Interface Sci.* 18 (3) (1963) 208–216.
- [4] M.F. Mei, B.M. Yu, M.Q. Zou, L.A. Luo, A numerical study on growth mechanism of dropwise condensation, *Int. J. Heat Mass Transf.* 54 (9–10) (2011) 2004–2013.
- [5] B.B. Mikic, On mechanism of dropwise condensation, *Int. J. Heat Mass Transf.* 12 (10) (1969) 1311–1323.
- [6] J.W. Rose, On mechanism of dropwise condensation, *Int. J. Heat Mass Transf.* 10 (6) (1967) 755–756.
- [7] A. Umur, P. Griffith, Mechanism of dropwise condensation, *J. Heat Transfer* 87 (2) (1965) 275–282.
- [8] W.H. Wu, J.R. Maa, Heat transfer in dropwise condensation, *Chem. Eng. J.* 12 (3) (1976) 225–231.
- [9] C. Graham, P. Griffith, Drop size distributions and heat-transfer in dropwise condensation, *Int. J. Heat Mass Transf.* 16 (2) (1973) 337–346.
- [10] J.W. Rose, L. Glicksma, Dropwise condensation - distribution of drop sizes, *Int. J. Heat Mass Transf.* 16 (2) (1973) 411–425.
- [11] C. Dietz, K. Rykaczewski, A.G. Fedorov, Y. Joshi, Visualization of droplet departure on a superhydrophobic surface and implications to heat transfer enhancement during dropwise condensation, *Appl. Phys. Lett.* 97 (3) (2010).
- [12] A. Lee, M.W. Moon, H. Lim, W.D. Kim, H.Y. Kim, Water harvest via dewing, *Langmuir* 28 (27) (2012) 10183–10191.
- [13] A.I. ElSherbini, A.M. Jacobi, Retention forces and contact angles for critical liquid drops on non-horizontal surfaces, *J. Colloid Interface Sci.* 299 (issue 2) (2006) 841–849.
- [14] D. Quere, M.J. Azzopardi, L. Delattre, Drops at rest on a tilted plane, *Langmuir* 14 (8) (1998) 2213–2216.
- [15] V.D. Ta, Laser textured superhydrophobic surfaces and their applications for homogeneous spot deposition, *Appl. Surf. Sci.* 365 (2016) 153–159.
- [16] T.P. Koukoravas, A. Ghosh, P.S. Mahapatra, R. Ganguly, C.M. Megaridis, Spatially-selective cooling by liquid jet impinging orthogonally on a wettability-patterned surface, *Int. J. Heat Mass Transf.* 95 (2016) 142–152.
- [17] D. Orejon, A. Askounis, Y. Takata, D. Attinger, Dropwise condensation on multiscale bioinspired metallic surfaces with nanofeatures, *ACS Appl. Mater. Interfaces* 11 (27) (2019) 24735–24750.
- [18] A. Speidel, J.W. Murray, I. Bisterov, J. Mitchell-Smith, C. Parmenter, A.T. Clare, Thermal activation of electrochemical seed surfaces for selective and tunable hydrophobic patterning, *ACS Appl. Mater. Interfaces* 12 (6) (2020) 7744–7759.
- [19] H. Jo, Loss of superhydrophobicity of hydrophobic micro/nano structures during condensation, *Sci. Rep.* 5 (2015).
- [20] J.Y. Long, Cassie-state stability of metallic superhydrophobic surfaces with various micro/nanostructures produced by a femtosecond laser, *Langmuir* 32 (4) (2016) 1065–1072.
- [21] X.H. Ma, S.F. Wang, Z. Lan, B.L. Peng, H.B. Ma, P. Cheng, Wetting mode evolution of steam dropwise condensation on superhydrophobic surface in the presence of noncondensable gas, *J. Heat Trans. T. ASME* 134 (2) (2012).
- [22] K.S. Lau, J. Bico, B.K. Teo, Superhydrophobic carbon nanotube forests, *Nano Lett.* 3 (12) (2003) 1701–1705.
- [23] R.D. Narhe, D.A. Beysens, Growth dynamics of water drops on a square-pattern rough hydrophobic surface, *Langmuir* 23 (12) (2007) 6486–6489.
- [24] C.H. Chen, C.Tsai Q.Cai, C.L. Chen, Dropwise condensation on superhydrophobic surfaces with two-tier roughness, *Phys. Lett.* 90 (2007).
- [25] C. Dorrier, J. Ruhe, Wetting of silicon nanoglass: from superhydrophilic to superhydrophobic surfaces, *Adv. Matt.* 20 (2008) 159–163.
- [26] A. Chatterjee, M.M. Derby, Y. Peles, M.K. Jensen, Condensation heat transfer on patterned surfaces, *Int. J. Heat Mass Transf.* 66 (2013) 889–897.
- [27] A. Chatterjee, M.M. Derby, Y. Peles, M.K. Jensen, Enhancement of condensation heat transfer with patterned surfaces, *Int. J. Heat Mass Transf.* 71 (2014) 675–681.
- [28] S. Daniel, M.K. Chaudhury, J.C. Chen, Fast drop movements resulting from the phase change on a gradient surface, *Science* 291 (5504) (2001) 633–636.
- [29] M.H.M. Grooten, C.W.M. van der Geld, Surface property effects on dropwise condensation heat transfer from flowing air-steam mixtures to promote drainage, *Int. J. Therm. Sci.* 54 (2012) 220–229.
- [30] K.S. Yang, K.H. Lin, C.W. Tu, Y.Z. He, C.C. Wang, Experimental investigation of moist air condensation on hydrophilic, hydrophobic, superhydrophilic, and hybrid hydrophobic-hydrophilic surfaces, *Int. J. Heat Mass Transf.* 115 (2017) 1032–1041.
- [31] B.L. Peng, X.H. Ma, Z. Lan, W. Xu, R.F. Wen, Analysis of condensation heat transfer enhancement with dropwise-filmwise hybrid surface: droplet sizes effect, *Int. J. Heat Mass Transf.* 77 (2014) 785–794.
- [32] B.L. Peng, X.H. Ma, Z. Lan, W. Xu, R.F. Wen, Experimental investigation on steam condensation heat transfer enhancement with vertically patterned hydrophobic-hydrophilic hybrid surfaces, *Int. J. Heat Mass Transf.* 83 (2015) 27–38.
- [33] P.S. Mahapatra, A. Ghosh, R. Ganguly, C.M. Megaridis, Key design and operating parameters for enhancing dropwise condensation through wettability patterning, *Int. J. Heat Mass Transf.* 92 (2016) 877–883.
- [34] C.V. Ngo, D.M. Chun, Control of laser-ablated aluminum surface wettability to superhydrophobic or superhydrophilic through simple heat treatment or water boiling post-processing, *Appl. Surf. Sci.* 435 (2018) 974–982.
- [35] Y. Jin, Atmospheric water harvesting: role of surface wettability and edge effect, *Global Challenges* 1 (4) (2017).
- [36] S.W. Churchill, Comprehensive correlating equation for forced-convection from flat plates, *AIChE J.* 22 (2) (1976) 264–268.
- [37] J.W. Rose, Boundary-layer flow on a flat-plate, *Int. J. Heat Mass Transf.* 22 (6) (1979) 969–969.
- [38] M.G. Medici, A. Mongruel, L. Royon, D. Beysens, Edge effects on water droplet condensation, *Phys. Rev. E* 90 (issue 6) (2014).
- [39] P.B. Bintein, H. Lhuissier, A. Mongruel, L. Royon, D. Beysens, Grooves Accelerate Dew Shedding, *Phys. Rev. Lett.* 122 (issue 9) (2019).